

HOMEWORK 1

35 points

1. Which of the following are *mathematical statements*? **4*1 = 4 points**

a) $2+3=5$

b) $x+2=7$

c) "Paris is the capital of France."

d) $4 < y + 2$

2. Let P: "7 is a prime number" and Q: "2 is an even number." **$1 + 1 = 2$ points**

- Write the conjunction $P \wedge Q$
- Is it true or false?

3. Let P: "5 is divisible by 2" and Q: "9 is greater than 7." **$1 + 1 = 2$ points**

- Write $P \vee Q$
- Determine its truth value.

4. Write the following statement using mathematical notations:

“Some integers are negative.”

Negate the above statement. Use mathematical notations in your final answer.

2 points

5. Let P be "4 is an odd number," Q: "10 is greater than 7," and R: "15 is divisible by 5." Find the truth value of the compound statement: $(\neg P \wedge Q) \vee R$. **3 points**

Show your steps and reasoning

6. Express the following in words: **2 + 2 = 4 points**

- $\forall x \in \mathbb{N}. \exists y \in \mathbb{N}. (y > x)$
- $\exists x \in \mathbb{R}. \forall y \in \mathbb{R}. (x + y = 0)$

7. Determine the truth value of the given statements: **3 * 1 = 3 points**

- $\forall x \in \mathbb{N}. \exists y \in \mathbb{N}. (x < y)$
- $\exists x \in \mathbb{N}. \forall y \in \mathbb{N}. (x < y)$
- $\exists x \in \mathbb{R}. (x^2 + 1 = 0)$

8. Construct Truth Tables for: **3 + 3 = 6 points**

a) $(P \wedge Q) \vee \neg R$

b) $(P \Rightarrow Q) \wedge (Q \Rightarrow R)$

9. Consider the statement: “if a number is divisible by 4, it is even” **3 + 3 = 6**
points

- Write the converse, inverse and contrapositive of the above statement
- Which of the above are True and which are False?

10. Determine the truth value of each statement (True/False), assuming the universe is all integers: **3 points**

- If n is divisible by 6, then n is divisible by 3.
- If n is divisible by 3, then n is divisible by 6.
- A number is divisible by 6 if and only if it is divisible by both 2 and 3.